Stage 1 Report: Implementation and Preliminary Verification of a Q4 Element in STAP++

Finite Element Method Course Project I

May 9, 2026

1 Purpose and Scope

This stage implements the minimum working loop for a four-node isoparametric quadrilateral element (Q4) in STAP++: input, material definition, stiffness assembly, solution, stress recovery, and preliminary verification. The full patch test, convergence study, ABAQUS comparison, and final course-project report will be completed in later stages.

2 Implemented Functionality

The new element class `CQ4` was added to support two-dimensional linear elasticity. Each Q4 element has four nodes and uses the first two nodal degrees of freedom, u_x and u_y. The third degree of freedom u_z must be constrained in Q4 examples to avoid unused zero-stiffness equations.

The material class `CQ4Material` uses the following input format:

```
nset E nu thickness mode
```

where `mode=1` denotes plane stress and `mode=2` denotes plane strain.

The following source-level changes were made:

- Added `src/h/Q4.h` and `src/cpp/Q4.cpp` for the Q4 element.
- Added `CQ4Material` in `Material.h/.cpp`.
- Connected `ElementTypes::Q4` in `ElementGroup.h/.cpp`.
- Added Q4 material, element, and stress output in `Outputter.h/.cpp`.
- Fixed two framework issues: `CElement` no longer deletes the non-owning material pointer, and `CElementGroup::Read` now checks material-read failures.

3 Element Formulation

The Q4 element uses the standard bilinear shape functions in natural coordinates (ξ, η):

$$N_1 = \frac{1}{4}(1-\xi)(1-\eta), \qquad N_2 = \frac{1}{4}(1+\xi)(1-\eta), \tag{1}$$

$$N_3 = \frac{1}{4}(1+\xi)(1+\eta), \qquad N_4 = \frac{1}{4}(1-\xi)(1+\eta). \tag{2}$$

The stiffness matrix is computed with a 2×2 Gauss rule:

$$K_e = \sum_{g=1}^{4} B_g^T D B_g \det(J_g) t. \tag{3}$$

Here B is the strain-displacement matrix, D is the plane-stress or plane-strain elasticity matrix, J is the isoparametric Jacobian, and t is thickness.

For plane stress,

$$D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}. \tag{4}$$

For plane strain,

$$D = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & (1-2\nu)/2 \end{bmatrix}. \tag{5}$$

Element stresses are currently reported at the element center $(\xi, \eta) = (0, 0)$ as $(\sigma_x, \sigma_y, \tau_{xy})$.

4 Input Examples

A typical Q4 element group uses element type 2:

```
2 NUME NUMMAT
nset E nu thickness mode
element_id node1 node2 node3 node4 material_set
```

The nodes should be ordered counter-clockwise. A non-positive Jacobian determinant is treated as an input error.

5 Preliminary Verification

The executable was rebuilt in a fresh build directory because the existing `build` directory pointed to an old source path. The successful commands were:

```
cmake -S src -B build-codex
cmake --build build-codex --config Debug
```

The original bar example and all new Q4 positive tests completed with exit code 0.

Table 1: Preliminary Q4 verification cases

Case	Result	Main observation
`q4-single.dat`	Passed	Plane-stress tension; right-edge $u_x = 4.62010 \times 10^{-3}$, $\sigma_x = 1.00000 \times 10^3$, $\tau_{xy} \approx 0$.
`q4-plane-strain.dat`	Passed	Plane-strain tension; right-edge $u_x = 4.05933 \times 10^{-3}$, smaller than plane stress as expected.
`q4-two-element.dat`	Passed	Two Q4 elements in series; right-edge $u_x = 9.37340 \times 10^{-3}$ and positive extension trend.
`q4-shear.dat`	Passed	Shear loading; recovered center stress $\tau_{xy} = 5.00000 \times 10^2$ with near-zero normal stress.

Two negative tests were also added. `q4-invalid-mode.dat` checks that invalid material mode is rejected. `q4-invalid-jacobian.dat` checks that clockwise or otherwise invalid node order is rejected through the Jacobian determinant check.

6 Current Limitations and Next Work

This stage only verifies that the Q4 implementation is functional and numerically reasonable on small examples. The following items remain for later completion:

- Formal patch test using an imposed linear displacement field.
- Mesh-convergence study using multiple grid densities.
- ABAQUS comparison for at least one representative plane-stress or plane-strain problem.
- Final report sections on convergence rate, verification figures, and course-project conclusions.